ASHA American Speech-Language-Hearing Association

JSLHR

Research Article

Is Pitch Height or Pitch Contour a Challenge? Production of Mandarin Tones in Hani–Mandarin Bilingual Children

Aliang Li[a,b]

[a] Zhejiang University, Hangzhou, China [b] École Pratique des Hautes Études-PSL, Paris, France

ARTICLE INFO

Article History:
Received November 16, 2024
Revision received March 24, 2025
Accepted March 28, 2025

Editor-in-Chief: Jessica E. Huber
Editor: Shaheen N. Awan

https://doi.org/10.1044/2025_JSLHR-24-00810

ABSTRACT

Purpose: The study aimed to determine the relative difficulty of Mandarin tone production in a classroom setting for native Hani-speaking children and to explore whether these bilingual children's tone production follows the same patterns observed in Mandarin monolingual children.
Method: This study investigates Mandarin tone production in Hani–Mandarin bilinguals (n = 30) compared to Mandarin monolinguals (n = 20) using Growth Curve Modeling to analyze differences in pitch height and pitch contour between second language (L2) and first language (L1) Mandarin children.
Results: Hani–Mandarin bilinguals produce Mandarin tones with distinct acoustic patterns compared to monolinguals: Hani–Mandarin bilingual children's Tone (T)1 and T4 differ from Mandarin monolingual children in pitch height, while T2 and T3 differ from Mandarin children in pitch contour. Hani–Mandarin bilinguals occasionally pronounce T3 as T2 or vice versa, displaying a bidirectional confusion between T2 and T3.
Conclusions: L1 experience significantly influences L2 tone production. Bilingual children with a tonal L1 produce tones with higher static pitch targets similar to monolingual children, regardless of the differences between the L1 and L2 tonal systems. Complex pitch contours and laryngeal muscle activation patterns in L2 tones may hinder tone acquisition for both bilingual and even native-speaking children.

Developing proficiency in segmental information (such as consonants and vowels) and suprasegmental information (including tone, stress, and intonation) is essential for the acquisition of second language (L2) phonology (Lee et al., 2019). Consequently, establishing contrastive categories that align with the bilingual input language presents a significant challenge in L2 learning. Suprasegmentals, particularly tone, play a critical role in distinguishing lexical meaning (Roll, 2015), but tonal contrasts can be particularly challenging for L2 learners. This is largely because the tonal systems of the two languages are unlikely to overlap. While several contrastive features are shared between the languages and exhibit similar distribution patterns, many others highlight differences in the division of acoustic space.

Over the past 4 decades, this phenomenon has been extensively studied in simultaneous bilinguals, as well as in early and late sequential bilingual children and adults, particularly regarding their abilities to produce tone contrasts. The present study investigates the production of four tone contrasts in Mandarin Chinese among Hani–Mandarin bilingual and Mandarin monolingual children. The goal is to gain a deeper understanding of how varying initial language experiences affect tonal production abilities. Specifically, this study examines the tone production accuracy of Hani–Mandarin bilingual children across the four Mandarin tones and provides a detailed acoustic comparison with Mandarin monolingual children.

Unlike previous studies that often center on the correctness of tone production alone, this study employs a

Correspondence to Aliang Li: lialiang0908@126.com. ***Disclosure:*** *The author has declared that no competing financial or nonfinancial interests existed at the time of publication.*

combined approach of human judgment and acoustic analysis. As Saito and Plonsky (2019) pointed out, acoustic information alone cannot fully simulate human speech perception. Therefore, to assess the proficiency of L2 learners in both segmental and suprasegmental features, it is crucial to integrate both human judgment and acoustic analysis, depending on the specific research objectives. This dual focus provides a more detailed analysis of which tonal aspects pose challenges for L2 learners. The insights derived from this approach could inform more effective methods to teaching Mandarin tones.

Background Literature

The Acquisition Theory of Bilingual or L2 Speech: Adults and Children

Theories of bilingual speech acquisition encompass both adult and child bilingualism, as well as L2 acquisition. To explain the challenges faced by L2 learners in the production of L2 vowels and consonants, Flege (1995) proposed the speech learning model (SLM), which outlines how individuals learn and produce L2 sounds. Flege argued that the accuracy of nonnative speech production is closely related to its perception; improved perception, he contended, leads to improved production. Furthermore, the SLM posits that the processes and mechanisms used by children to learn their native language remain consistent throughout the lifespan (Flege, 1999, 2003). However, this model was later revised into the revised SLM (SLM-r; Flege & Bohn, 2021). The SLM-r focuses on how sequential bilinguals—both early and late learners—produce and perceive L2 vowels and consonants, explaining how the phonetic system adapts over the lifespan based on speech input received during naturalistic learning. Additionally, the SLM-r challenges the dominance of L2 experience, the relationship between L2 category formation and age, and the connection between production and perception. According to the SLM-r (Flege & Bohn, 2021, p. 65), "the ability of an L2 sound to form a new phonetic category depends primarily on (1) the degree to which the sound is phonetically different from the closest L1 sound, (2) the amount and quality of L2 input that the sound receives in meaningful conversations, and (3) the precision with which the category of the closest L1 sound is assigned at the onset of L2 learning."

Bilingual children grow up learning or using two languages, and language acquisition theories in bilingual children focus on whether they develop one or two distinct language systems. One such theory, the unitary language system, posits that bilingual children exposed to two languages from birth may pass through a merged system stage during language development (Redlinger & Park, 1980; Schnitzer & Krasinski, 1994; Volterra & Taeschner, 1978). In contrast, the dual language system (DLS) suggests that bilingual children develop two separate language systems from the outset of language acquisition (Genesee, 1989; Johnson & Lancaster, 1998; Paradis, 2001; Johnson & Wilson, 2002; Keshavarz & Ingram, 2002; Schnitzer & Krasinski, 1996), meaning that bilingual children receive two linguistic inputs and separate them into distinct systems. The processing rich information from multidimensional interactive representations (PRIMIR) model (Curtin et al., 2011) also supports the DLS view, asserting that bilingual infants can effectively distinguish between their two languages by comparing their phonological features. With increased exposure to L2, the bilingual child's DLS continues to develop, with the two systems interacting in a pattern of interdependence (Fabiano-Smith & Barlow, 2010).

Paradis and Genesee (1996) suggest that bilingual children may develop separate language systems that are either autonomous, where the systems do not interact, or dependent, where the systems do interact. If the language systems develop autonomously, the acquisition of each language by bilinguals will resemble that of monolinguals. However, when the systems interact, three processes—migration, acceleration, or delay—can occur within each language. Interestingly, Grosjean (1997) directly argued that bilinguals possess two separate and independent phonological systems, with each system functioning similarly to that of a monolingual speaker. This implies that bilinguals use the corresponding phonological system in different linguistic contexts, thereby producing or perceiving speech information as a monolingual would in each respective language environment (MacLeod & Stoel-Gammon, 2005; Sundara et al., 2006).

The existing SLMs were not originally designed to address the cross-tier production relationships that may influence L2 learners' nonnative tone production. For L2 learners, a key question is whether and how the assimilation of lexical tones into native prosodic contrasts facilitates or hinders the acquisition of lexical tones in a tonal language. Another issue concerns whether and how cross-tier influences between production and perception differ in quantity and/or quality from nonnative tone contrasts at the same tier, as well as from reference contrasts in the native language (Best, 2019). Since these models were specifically developed to account for cross-language production and perception of nonnative consonants and vowels relative to native segments, careful consideration is required when attempting to extend existing SLMs—such as the SLM (Flege, 1995), the SLM-r (Flege & Bohn, 2021), and the PRIMIR (Curtin et al., 2011)—to studies on L2 learners' production and perception of nonnative lexical tones.

Undeniably, models such as SLM and SLM-r can be fairly directly extended to predict how L2 learners with

a tonal first language (L1) produce and distinguish nonnative tones at the segmental level. Additionally, a brief supplementary discussion by Curtin and Werker (2018) has highlighted certain L2 tone research findings that support the PRIMIR framework. Specifically, nonnative speakers' sensitivity to lexical tones is shaped by both general perceptual development and simultaneous access to multiple representational planes. Furthermore, when nonnative speakers access acoustic information embedded within the general perceptual plane, their native phonological system constrains the mapping between nonnative contrasts and distinct lexical items, making them more likely to treat lexical tones as phonemic elements. However, whether bilingual experience enhances or reduces bilinguals' attention to acoustic and phonological information in lexical tones, and which specific type of bilingual experience (e.g., tonal vs. nontonal) facilitates attention to tone, remain critical areas for future research.

Bilingual or L2 Speech Acquisition: Insights From Psycholinguistics and Neuroscience

Psycholinguists investigate the cognitive processes and mechanisms underlying language production, comprehension, acquisition, and usage. They argue that uncovering the structures and processes in learners' minds is central to understanding second language acquisition (SLA). In other words, SLA can be regarded as a subfield of psycholinguistics. Meanwhile, with the increasing number of bilingual individuals, bilingualism research and SLA have become focal areas in psycholinguistic studies (Prior & van Hell, 2021).

In the field of psycholinguistics, the Levelt model was initially developed to explain L1 speech production (Levelt, 1989). Subsequently, this model was extended to L2 speech production (e.g., de Bot, 1992; Kormos, 2006; Segalowitz, 2010), primarily to compare the differences and similarities between native and nonnative speakers' speech output and to describe various aspects of L2 phonetic production from a psycholinguistic perspective. According to de Bot (1992), phonological encoding is triggered by selecting lexical items from a subset of the mental lexicon. Regardless of how many languages a speaker has acquired, the articulatory system remains singular and independent, which accounts for the common phenomenon of L1-accented speech among L2 learners. Kormos (2006, 2011) further classified the knowledge required for L2 speech production into episodic memory, declarative memory, and semantic memory, where declarative memory is responsible for encoding L2 syntactic and phonological rules that are not proceduralized in L1. Notably, these revised models emphasize that L2 speech production primarily relies on working memory and attentional resources (Kormos, 2006; Segalowitz, 2010), whereas L1 speech production is generally considered subconscious and automatic (Levelt, 1989).

The role of working memory and attention in L2 speech production has been extensively investigated. Research suggests that both the executive aspect of working memory and attentional control contribute to enhanced success rates in L2 production and comprehension (see reviews by Linck et al., 2014; Wen et al., 2013). Since the executive aspect of working memory is conceptualized as an attentional control system, findings from these two measures often exhibit similar patterns (Simard et al., 2016). Studies have demonstrated that individuals with greater working memory spans perform better in L2 English phonological production tasks (Fortkamp, 1999). Moreover, L2 attention research has been widely conducted from the perspectives of attentional characteristics and attentional executive functions (see Kormos, 2011; Simard & Wong, 2001; Zuniga, 2015; Zuniga & Simard, 2019, for reviews). Among these studies, selective attention has been identified as a crucial factor in various aspects of L2 development (Kormos, 2006, 2011). Overall, psycholinguistic research on L2 speech production highlights the fundamental role of working memory and attention in SLA. Therefore, adequate production time and repeated task practice are essential in L2 phonetic learning.

VanPatten (2014) argues that neurolinguistics and its methodologies are increasingly being applied to investigate language processing and that any introduction to SLA should include neurolinguistics and its research methods. He further states that the primary focus of neurolinguistic research in SLA is to examine neural activity (e.g., electrical impulses, blood flow) to determine the extent of convergence or divergence between L1 and L2 processing. Event-related potentials (ERPs) have been among the most widely used techniques in SLA research. However, findings from ERP studies have been mixed regarding whether L2 learners can achieve native-like processing. Some studies suggest that L2 learners can process language similarly to native speakers (Osterhout et al., 2006), while others argue that native-like processing is elusive or, at best, highly challenging (Mueller, 2006). The discrepancy in findings may be associated with learning methods, as explicit learning does not yield the same consistent native-like processing effects as implicit learning (Morgan-Short et al., 2012; VanPatten et al., 2012).

In addition to ERP, functional magnetic resonance imaging has also been employed in neurolinguistic research to investigate the neural mechanisms underlying L2 acquisition, specifically comparing the extent to which monolinguals and bilinguals engage similar or distinct brain regions for language processing. Compared to L1 speakers, L2 learners appear to recruit different brain

regions for language processing (Kotz, 2009). However, some studies (e.g., Abutalebi et al., 2009) suggest that L2 learners do not activate entirely different brain regions compared to native speakers but instead engage additional brain areas beyond those used by L1 speakers.

Neurolinguistic research on lexical tone processing indicates that lexical tone processing is associated with both subcortical and cortical structures (Gandour & Krishnan, 2016). In early L2 acquisition, lexical tone processing is primarily handled by the right bilaterally or hemisphere. However, as language exposure increases, processing gradually lateralizes to the left hemisphere, aligning with native-like patterns (Gandour et al., 2004; Kaan et al., 2013; Wang et al., 2004).

Pitch Height and Pitch Contour

Tone can generally be characterized by two acoustic parameters: pitch height and pitch contour. Simply put, "pitch height refers to the exact placement of the pitch of a syllable within a given pitch span, while pitch contour represents the shape of the pitch value within the duration of the syllable bearing it" (Trask, 1995, p. 279). Notably, pitch height in this context pertains to a physical acoustic concept rather than a perceptual category, as it is measured in Hertz (Hz) and corresponds to the fundamental frequency (*F*0) of the signal. Given the well-established psychophysical relationship between pitch perception and the *F*0 of a periodic sound source (Beckman & Venditti, 2010, p. 604), some studies have conceptualized pitch height as a psychoacoustic construct by transforming its physical representation into a perceptual one. Anand et al. (2021, p. 185) define the *F*0 of the candidate waveform, which exhibits the highest degree of similarity (correlation between 0 and 1) between the processed sawtooth waveform and the identically processed spectrum of the input signal, as the estimated pitch height. Despite such transformations aligning pitch height more closely with perception, it remains an acoustic parameter expressed in *F*0 (Hz). Over a given time span, pitch height may either change dynamically (rising or falling) or remain stable, thereby forming distinct pitch patterns. In other words, pitch height partially determines pitch contour. Consequently, pitch contour can manifest as rising or falling shapes, or as more complex configurations combining the two, such as fall–rise and rise–fall patterns.

Languages with lexical tones utilize variations in pitch height and pitch contours to differentiate word meanings (Rattanasone et al., 2018). Mandarin Chinese is widely recognized as a tone language, and its tonal characteristics have long been a subject of investigation using modern experimental methodologies. Chao (1922) pioneered the systematic study of Mandarin tones by employing an experimental apparatus equipped with flutes, which enabled him to document pitch variations on a musical staff. Through this method, he identified four distinct pitch contours: high-level contour, high-rising contour, low falling–rising contour, and high-falling contour. Expanding on this foundation, Chao (1933) proposed a 5-point scale notation system to encode pitch height in Mandarin. In this system, pitch height is ranked from 1 (lowest) to 5 (highest), and each tone is described based on its initial and final pitch levels. However, this notation primarily reflects a perceptual outcome rather than the actual pitch of the tone.

In the process of L2 tonal acquisition, the primary cue for distinguishing tonal categories in an L2 is *F*0, with L2 learners relying on differences in pitch height and pitch contour to establish tonal distinctions (Pelzl et al., 2022). Our study adopts the conceptual framework outlined by Gandour (1979, 1983) to guide our investigation. According to Gandour (1979, 1983), pitch height corresponds to the mean *F*0, while pitch contour is represented by its slope. Within this framework, both mean pitch height and contour slope are linked to the characteristics of *F*0, where slope encompasses both level and contour distinctions. This model aligns well with the acoustic realization of Mandarin tones, wherein pitch height specifically refers to the actual *F*0 of a tone, while pitch contour captures the static or dynamic changes in *F*0 over a given time span. Consequently, these parameters serve as two key acoustic dimensions for examining differences in Mandarin tone production between native speakers and L2 learners.

Tonal Systems in Mandarin and Hani

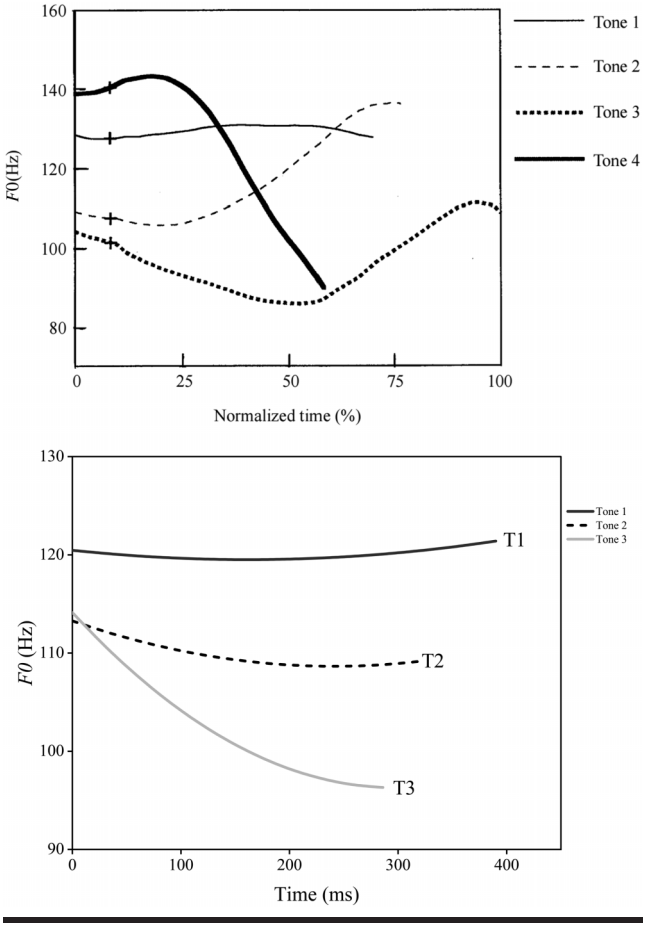
Mandarin is a tonal language characterized by four lexical tones, which serve to distinguish between different lexical meanings. Tone 1 (T1) is high level (55), Tone 2 (T2) is high rising (35), Tone 3 (T3) is low falling–rising (214), and Tone 4 (T4) is high falling (51). For instance, 妈 (mā) means "mother" in T1, 麻 (má) means "hemp" in T2, 马 (mǎ) means "horse" in T3, and 骂 (mà) means "scold" in T4. These four tones in Mandarin differ in pitch onset, contour, and duration (Li et al., 2017; Liu & Samuel, 2004; Xu, 1997). Hani is also classified as a tonal language, though studies vary regarding the number of tonal categories, with estimates ranging from three to five (Bradley, 1977; Edmondson, 2002; Paul & Bai, 1996; Wheatley, 1982). Despite this disagreement, three tones are widely recognized: high level (55), middle level (33), and low falling (31). The distinction between three and five tones is actually related to phonation type. Specifically, in Hani, vowels are categorized into two sets: "lax" and "tense," which correspond to modal voice and creaky voice, respectively. The middle level (33) and low falling (31) tones are further divided based on their combination with different vowel types. When these tones occur with

lax vowels, they are realized as lax middle level (33) and lax low falling (31). Conversely, when they occur with tense vowels, they are realized as tense middle level (33) and tense low falling (31). This results in a discrepancy in the number of tones between Hani and Mandarin—three versus four. While Mandarin includes a rising tone, Hani lacks this feature, as illustrated in Figure 1. Consequently, it is challenging for Hani–Mandarin bilinguals to master the Mandarin-specific rising tones, as these tones are likely to be assimilated into a single category in Hani (Yang, 2018).

The Production of Mandarin Tones in Monolingual and Bilingual Children

Compared to research on how children perceive Mandarin tones, there is very limited research on the production of Mandarin tones. These studies mainly focus on monolingual children, and there are very few studies on bilingual children. Early studies focused on the accuracy of Mandarin tone production (Chao, 1973; Hua, 2002; Hua & Dodd, 2000) and the age or time of acquisition (Clumeck, 1977; Li & Thompson, 1977) of monolingual Mandarin children. According to Li and Thompson (1977), children acquire the four tones of Mandarin in a relatively short period, that is, they master the four tones of Mandarin when they begin to produce sentences of more than two to three words. However, some studies point to individual differences in children's acquisition of Mandarin, and two monolingual children aged 32 and 34 months were found to have not yet fully acquired Mandarin tones (Clumeck, 1977). Based on these studies, later studies have turned to the comparison of Mandarin tones produced by Mandarin children and Mandarin adults. Many studies have shown differences in Mandarin tone production between children and adults, including for monosyllabic words (Wong et al., 2005) and for disyllabic words (Wong, 2008). Wong (2012) found that although adults judged the four tones produced by 3-year-olds to be correct, they were acoustically different from those of adults. Specifically, the production of T1 and T3 by children differs from that of adults in both pitch height and pitch contour. Compared to adults, children's T1 is neither as level nor as high, and their T3 exhibits a higher pitch height with a contour that is not as low as that of adults. Additionally, the *F*0 range of children's T2 is significantly smaller than that of adults. She also found that children tended to produce high-frequency tones while they had more difficulty producing low-frequency tones. Not only the *F*0 of the tone but also the duration of the tone is acoustically different between children and adults (Plant & Ge, 2010). Yang et al. (2021) found that the duration of the four tones produced by children of different ages was different. Although 5-year-old and 8-year-old children both showed similar durations to adults, 5-year-old children did not show differences in the duration of T2 and T3, while 8-year-old children showed significant differences. In the production of two-syllable tones, children tend to shift the *F*0 of the first syllable to reduce *F*0 variation, and they show more anticipatory coarticulation than adults (Wong & Strange, 2017). Rattanasone et al. (2018) found that 3- to 5-year-old Mandarin children produced adultlike global contours in monosyllabic and disyllabic tones, but they were still fine-tuning their pitch range, slope, and curvature.

Figure 1. The top panel displays the mean *F*0 contours of the four Mandarin tones in the monosyllable /ma/ produced by eight male Mandarin speakers in isolation (Xu, 1997). The bottom panel shows the mean *F*0 contours of the three Hani tones produced by a Hani speaker in isolation (the acoustic data of Hani language were collected by the authors of this article). *F*0 = fundamental frequency. Top panel reprinted from *Journal of Phonetics, 25,* Xu, "Contextual Tonal Variations in Mandarin," 61–83, Copyright © 1997, with permission from Elsevier.



The production of Mandarin tones in bilingual children is not as well researched as in monolingual children, and there are not as many perceptual studies on the production of Mandarin tones in bilingual children. In the studies on the production of Mandarin tones in bilingual children, the children's language background includes both tonal and nontonal languages. These studies mainly analyze

the mapping relationship between children's native language and L2 Mandarin tones (Huang, 2016; You et al., 2024) and the acoustic differences between Mandarin tones of monolingual children and bilingual children (Fabra et al., 2021; Lin et al., 2025; Meckelborg et al., 2024). Huang (2016) found that Min-Mandarin bilingual children can assimilate Mandarin's T1, T3, and T4 to the high level, low falling, and high falling tones of their native language. However, the tone sandhi rules of their native language affect the production of Mandarin T2. This discrepancy can be attributed to the phonological constraints of Southern Min, where rising tones do not occur in non–XP-final positions. Consequently, Min-Mandarin bilingual children tend to assume that rising tones cannot appear in non–XP-final positions, leading them to produce a flatter Mandarin T2 instead of the correct rising tone in such contexts. This context-dependent variation stems from L1 phonological rules rather than physiological constraints on articulation. Notably, 6- and 7-year-old Fuzhou-Mandarin bilingual children, who also speak a variety of Min, produced Mandarin tones with high accuracy, closely resembling those of adults (You et al., 2024). Unlike the previous studies, Meckelborg et al. (2024) included three children from different native language backgrounds (Mandarin, English, and Cantonese) to examine the effects of tonal (Cantonese or Mandarin) and nontonal (English) experience on the acquisition of Mandarin tones. This study found that Cantonese–Mandarin bilingual children had comparable accuracy rates to Mandarin monolingual children. Cantonese-dominant bilingual children had more accurate tone production than English-dominant bilingual children. Although the bilingual children in these studies had different backgrounds, it seems that children with tonal experience are generally more likely to produce Mandarin tones similar to monolingual children. This may be due to the fact that the children's L1 is a dialect of Chinese that is closer to Mandarin. Three Cantonese tones (T1, T2, and T4) share similar pitch contours with Mandarin T1, T2, and T4, corresponding to high-level, high-rising, and falling contours, respectively. This further suggests that a tonal language background facilitates positive transfer in Cantonese–Mandarin bilingual children during the early stages of L2 Mandarin acquisition. In addition, early home language experience and recent school learning experience interact with the target tones and have different effects on the tones of bilingual children (Lin et al., 2025).

The Present Study

Research on consonants and vowels has revealed diverse trends in language acquisition, yet studies on suprasegmental features, such as tones, remain less frequent despite their critical importance (Fikkert et al., 2020). This underscores the need for further investigation into the acquisition of suprasegmental features. Consequently, the present study focuses on tone acquisition. As discussed in the literature review, while L2 tone perception has been extensively studied, research on L2 tone production remains noticeably limited. Meanwhile, studies comparing tone production between monolingual and bilingual children in terms of both pitch height and pitch contour remain particularly scarce. This study aims to fill this gap by focusing on L2 tone production, specifically examining Hani–Mandarin bilingual children as the target participants. The Hani ethnic group is the second-largest minority group in Yunnan Province, China, with a population exceeding 1 million. However, there is a lack of research on how Hani children acquire Mandarin. This study investigates the differences in acquiring Mandarin monosyllabic tones between Hani–Mandarin bilingual children and Mandarin monolingual children. By comparing the tone production of Hani–Mandarin bilinguals with that of Mandarin monolinguals, this research aims to provide a more precise analysis of the acoustic differences between L2 learners and native speakers. Such findings will contribute to enhancing pedagogical approaches for Mandarin instructors teaching Hani-speaking students.

Within language acquisition research, bilingualism is defined in three primary ways: "(1) exposure to two languages from birth, (2) regular use of more than one language in daily life, and (3) a continuum of language use and proficiency" (Grech & McLeod, 2012, p. 121). Some children experience bilingual exposure from birth, known as simultaneous bilingualism. Others, who are introduced to an L2 after their initial language, are termed sequential bilinguals (Serratrice, 2018). Most bilingual participants in our study were exposed to Mandarin at ages 2–6 years, making them sequential bilinguals. Although Hani was the primary language used at home, these children attended schools where Mandarin was the primary instructional language, following the immersion program guidelines used in public schools. Mandarin instruction formally begins at ages 5–6 years, with elementary school completed by ages 10–12 years. Accordingly, we selected Hani–Mandarin bilingual children aged 10–12 years nearing the end of elementary school, by which time they had received at least 6 years of formal Mandarin education.

While many studies have explored the acquisition of L2 suprasegmental features, research specifically examining Mandarin tone production in classroom settings is still emerging. This study uniquely investigates how native Hani speakers process L2 Mandarin tones in an educational context. To our knowledge, no published study has focused on the production of unfamiliar, nonnative tones by native Hani-speaking children. This research provides valuable insights into how speakers of a distinct and underresearched tonal language, such as Hani, process Mandarin tonal categories.

The study aimed to determine the relative difficulty of Mandarin tone production in a classroom setting for native Hani-speaking children and to explore whether these bilingual children's tone production follows the same patterns observed in Mandarin monolingual children. Specifically, we addressed the following research questions:

RQ 1. Which Mandarin tones are more difficult for Hani–Mandarin bilingual children?

We hypothesize that T2 (rising tone) and T3 (low falling–rising tone) will be more challenging to master than T1 (level tone) and T4 (falling tone), as the rising tone do not exist in the Hani tonal system. Articulation accuracy depends on the demand for articulatory movement, in which case T1 and T4 are easier to produce, similar to Mandarin monolingual children.

RQ 2. Which aspect of Mandarin tones is more challenging to master? Pitch height or pitch contour?

We hypothesize that Hani–Mandarin bilingual children will approximate the pitch heights of Mandarin tones based on their native tonal experience. However, we anticipate that the pitch contours of the two rising tones will differ markedly from those produced by Mandarin monolingual children.

Method

Participants

This study included two groups of sixth-grade students from Yunnan Province, China, with a total of 50 participants: experimental and control groups. Participants were 10- to 12-year-olds from either Hani- or Mandarin-speaking families, attending public schools where Mandarin was the primary language of instruction. The experimental group comprised 30 native Hani-speaking children (24 male participants) who learned Mandarin as L2. These participants, born in China to Hani-speaking parents, used only Hani at home. The control group included 20 native Mandarin children (11 male participants) who used Mandarin exclusively at home and school.

To differentiate language backgrounds, a questionnaire was administered to both groups. For the Hani-speaking bilingual children, questions focused on the age they began learning Mandarin and a self-assessment of their Mandarin proficiency. For the monolingual Mandarin-speaking group, the questionnaire addressed their exposure to other languages and included a self-rating of their Mandarin proficiency. The bilingual children had consistent exposure to Mandarin by the time of the study, while the Mandarin monolinguals had no L2 exposure and spoke Mandarin from birth. Appendix A summarizes participants' biographical details and Mandarin proficiency. The study was approved by the local ethics committee under general ethics approval, and all children and their parents provided informed consent for the study in accordance with the Declaration of Helsinki.

Speech Materials and Recording Procedure

The target words for this study were designed using three vowels, [a], [i], and [u], combined with two unaspirated voiceless stop consonants, [p] and [t]. This resulted in six syllable combinations: [pi], [pu], [pa], [ti], [tu], and [ta], all of which followed a consonant–vowel structure. In total, 24 target words were generated (see Appendix B). Each participant was recorded individually using a Sony ICD-UX560 digital recorder with a built-in microphone in a quiet room at their school. Both Hani–Mandarin bilingual and Mandarin monolingual participants read each target word within the same carrier sentence: "我现在读的是_ _ _ _这个词" ("I am reading the word _ _ _ _"). Before recording, participants reviewed the word lists to ensure familiarity. During the recording session, they were instructed to read each word at a standard pace, repeating each word twice. If a participant mispronounced a word or displayed significant disfluency, they were asked to reread the word for accuracy.

Data Measure and Analysis

The study collected a total of 2,400 speech samples, with Hani–Mandarin bilingual children (24 words × 2 repetitions × 30 speakers) producing 1,440 samples and Mandarin monolingual children (24 words × 2 repetitions × 20 speakers) providing 960 samples.

Phonetic transcription. To address the first research question, phonetic transcription was used to analyze overall production accuracy. Each audio file was independently assessed by two native Mandarin-speaking researchers trained in phonetics, who transcribed the tones produced in each recording. The raters listened to the recordings separately in a quiet room while wearing headphones. Each tone was categorized as one of the four Mandarin citation tones based on perceptual judgment, without reference to the other rater's decision.

Production accuracy analysis. Tone production accuracy can generally be determined using either a strict criterion or a lenient criterion, both of which are considered acceptable in tonal research (Mok & Lee, 2018). Specifically, under the lenient criterion, a tone was considered correct if at least one rater identified it as matching the expected tone. In contrast, under the strict criterion, a tone was only considered correctly produced if both raters independently classified it as the expected tone. This study adopted the strict criterion for analysis. To ensure the reliability of the ratings, interrater reliability was

calculated for different data sets. The results showed that the interrater reliability was high for both Hani–Mandarin bilingual children (Cronbach's α = .951) and Mandarin monolingual children (Cronbach's α = .973), both exceeding 90%, indicating strong agreement between raters.

To address the second research question, the acoustic characteristics of pitch height and pitch contour were compared between Hani–Mandarin bilingual and Mandarin monolingual children. This process involved three steps: acoustic measurement, *F*0 normalization, and statistical analysis of normalized *F*0.

Acoustic measurement. To analyze the recordings, the Textgrid annotation tool in the Praat software (Boersma & Weenink, 2022) was used to manually label the vowel portions of each target word. A Praat script then automatically extracted the *F*0 values at 10 evenly spaced time points across each vowel.

*F*0 *normalization*. To standardize the data, we applied a logarithmic *z*-score (LZ) transformation, a method proven effective for pitch normalization (e.g., Rose, 1987; Zhu, 1999). This normalization process minimizes individual differences by filtering out speaker-specific characteristics, allowing for a consistent comparison of pronunciation styles. The LZ normalization steps included calculating the *F*0 mean, converting it to a logarithmic scale, finding the logarithmic mean and standard deviation, and applying *z*-score normalization. The formula for this transformation is as follows:

$$Z = (x_i - \mu)/\sigma \quad (1)$$

where "x_i" represents the logarithmic value of the *F*0 mean at any measurement point, "μ" is the logarithmic value means, and "σ" is the standard deviation.

Statistical modeling of pitch contours and heights. To model the pitch contours and heights of the four Mandarin tones, a growth curve analysis (Mirman, 2014) was applied using quadratic time polynomial models, an approach demonstrated to be effective for Mandarin tone analysis (Chen et al., 2017). Our analysis included three components: intercept, linear time polynomial, and quadratic time polynomial. The intercept represents the *F*0 values across all measured time points, where a higher estimate indicates a higher pitch height. The linear time polynomial reflects the steepness of the slope, with a larger estimate suggesting a steeper incline, meaning a greater difference between *F*0 onset and offset, which corresponds to a wider tonal range. The quadratic time polynomial captures the curvature of the pitch contour, with a higher estimate indicating greater concavity or convexity, shaping the contour into a more pronounced U shape or ∩ shape. This model compares the pitch contours of Hani–Mandarin bilingual children with those of Mandarin monolingual children. The modeling formula used (Mirman et al., 2008) is as follows:

$$Y_{ij} = (\gamma_{00} + \zeta_{0i}) + (\gamma_{10} + \zeta_{1i}) \times \text{Time}_{ij} + \varepsilon_{ij}. \quad (2)$$

The formula used to model the extracted *F*0 value is as follows: "*i*" represents the *i*th *F*0 contour and "*j*" is the *j*th time point. The population means of the intercept and slope are represented by "γ_{00}" and "γ_{10}," respectively. The variability of an individual's intercept and slope are modeled by "ζ_{0i}" and "ζ_{1i}," respectively. The error terms are represented by "ε_{ij}." "Orthogonal polynomials were used to avoid any correlation of the linear and quadratic terms" (Mirman, 2014, p. 52). In the study, the Hani–Mandarin bilingual children and Mandarin monolingual children *F*0 contours for each tone were compared by performing a likelihood ratio test on a model in which they were treated as the same and on a model in which they were treated as different. A distinction between the members of each pair of surface *F*0 contours is indicated by a significant difference in the results of the model comparison. The model included *F*0 measured at 10 time points as the dependent variable, with child group as a fixed factor. Each participant was included as a random variable, with random intercepts estimated separately for each child group.

Results

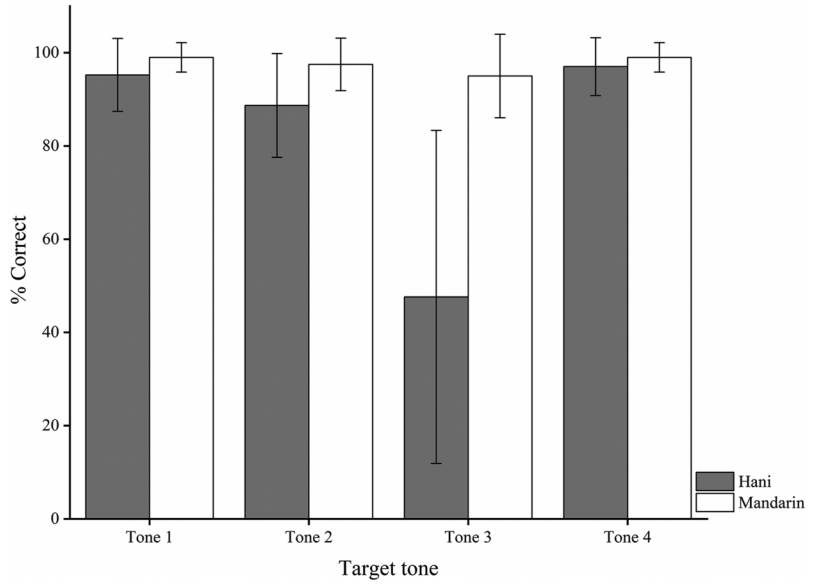
Overall Production Accuracy

Under the strict scoring criterion, the average tone production accuracy for Hani–Mandarin bilingual children and Mandarin monolingual children was 85.12% and 97.33%, respectively. These results indicate a significant discrepancy in proficiency levels between the two groups. Although both groups demonstrated high average accuracy, the accuracy of Hani–Mandarin bilingual children was considerably lower than that of Mandarin monolingual children.

Figure 2 illustrates the accuracy of individual tone production for both Hani–Mandarin bilingual and Mandarin monolingual children under the strict criterion. Across both groups, T2 and T3 exhibited lower accuracy compared to T1 and T4, indicating a developmental lag in these two tones. When comparing bilingual and monolingual children, bilinguals performed less accurately in producing T1, T2, and T3, while their accuracy in T4 production was the closest to that of monolingual children. This explains the substantial overall accuracy gap between the two groups.

Notably, the error bars for bilingual children were generally longer, particularly for T3, suggesting greater

Figure 2. Interspeaker normalized contours for Mandarin Chinese T1 (level), T2 (high rising), T3 (low falling rising), and T4 (high falling) produced by the Hani–Mandarin bilingual and Mandarin monolingual children. Numbers 1–10 indicate the 10 normalized time *F*0 measurements of the contours. The solid and dotted lines represent the fitted values of the growth curve models. *F*0 = fundamental frequency.



individual variation within the bilingual group compared to the monolingual group. Additionally, a mismatch pattern analysis was conducted (see Appendix C for confusion matrix results). The results revealed a T2 → T3 mismatch among Hani–Mandarin bilinguals. Moreover, T3 → T2 mismatch pattern was also dominant within their T3 production, indicating a bidirectional confusion between T2 and T3 among Hani–Mandarin bilingual children. Therefore, we compared the growth curves of T2 and T3 in the speech production of Hani–Mandarin bilingual children. The statistical results show that the quadratic term, $\chi^2(1) = 11.564$, $p < .001$, and the slope, $\chi^2(1) = 11.345$, $p < .001$, for T2 and T3 of Hani–Mandarin bilingual children were significant, but not the intercept, $\chi^2(1) = 3.20$, $p = .074$. This suggests that the Hani–Mandarin bilingual children could not distinguish Mandarin T2 and T3 in pitch height, although they could produce Mandarin T2 and T3 with different pitch contours.

Pitch Heights and Contours

The results of the growth curve analysis, presented in Table 1, reveal significant differences in the *F*0 contours of all four Mandarin tones between Hani–Mandarin bilingual children and native Mandarin-speaking children. A summary of the final model for each tone can be found in Appendix D.

Figure 3a displays the T1 contour patterns produced by Hani–Mandarin bilingual children and Mandarin monolingual children. When looking at the pitch contours, there are apparent cross-linguistic differences between the two groups. The Hani–Mandarin bilingual children largely succeeded in modeling the pitch contours of their Mandarin-speaking peers, with both groups producing more static contours. However, the Hani–Mandarin bilingual children were somewhat lower in pitch height than the Mandarin monolingual children. The main differences were in the onset and offset points of the pitch contour. Statistical comparison of the contours produced by the Hani–Mandarin bilingual children and the Mandarin monolingual children using growth curve analysis revealed a significant difference in the intercept, $\chi^2(1) = 4.274$, $p = .039$, indicating that there is a significant difference in the *F*0 heights of this pitch produced by the Hani–Mandarin bilingual children and the Mandarin children. However, the growth curve analysis does not yield significant differences in the slope in the linear, $\chi^2(1) = 1.689$, $p = .194$, and quadratic terms, $\chi^2(1) = 3.195$, $p = .074$. These results suggest that this tone's pitch contour was no different between the two groups.

Table 1. Growth curve analysis results for the four Mandarin tones.

	Quadratic term		Slope		Intercept	
Tone	$\chi^2(1)$	*p* value	$\chi^2(1)$	*p* value	$\chi^2(1)$	*p* value
Tone 1	3.195	.074	1.689	.194	4.275	.038*
Tone 2	31.715	< .001***	5.149	.023*	1.631	.202
Tone 3	23.820	< .001***	1.343	.247	2.867	.090
Tone 4	.895	.344	1.947	.163	9.537	.002**

*$p < .05$. **$p < .01$. ***$p < .001$.

Figure 3. A schematic illustration of the hypothetical surface *F*0 realization (solid line) and pitch target (dotted lines): (a) the hypothetical surface *F*0 (solid curved line) and a single static target (dotted line); (b) the hypothetical surface *F*0 (solid curved line) and a single dynamic target (dotted line). *F*0 = fundamental frequency.

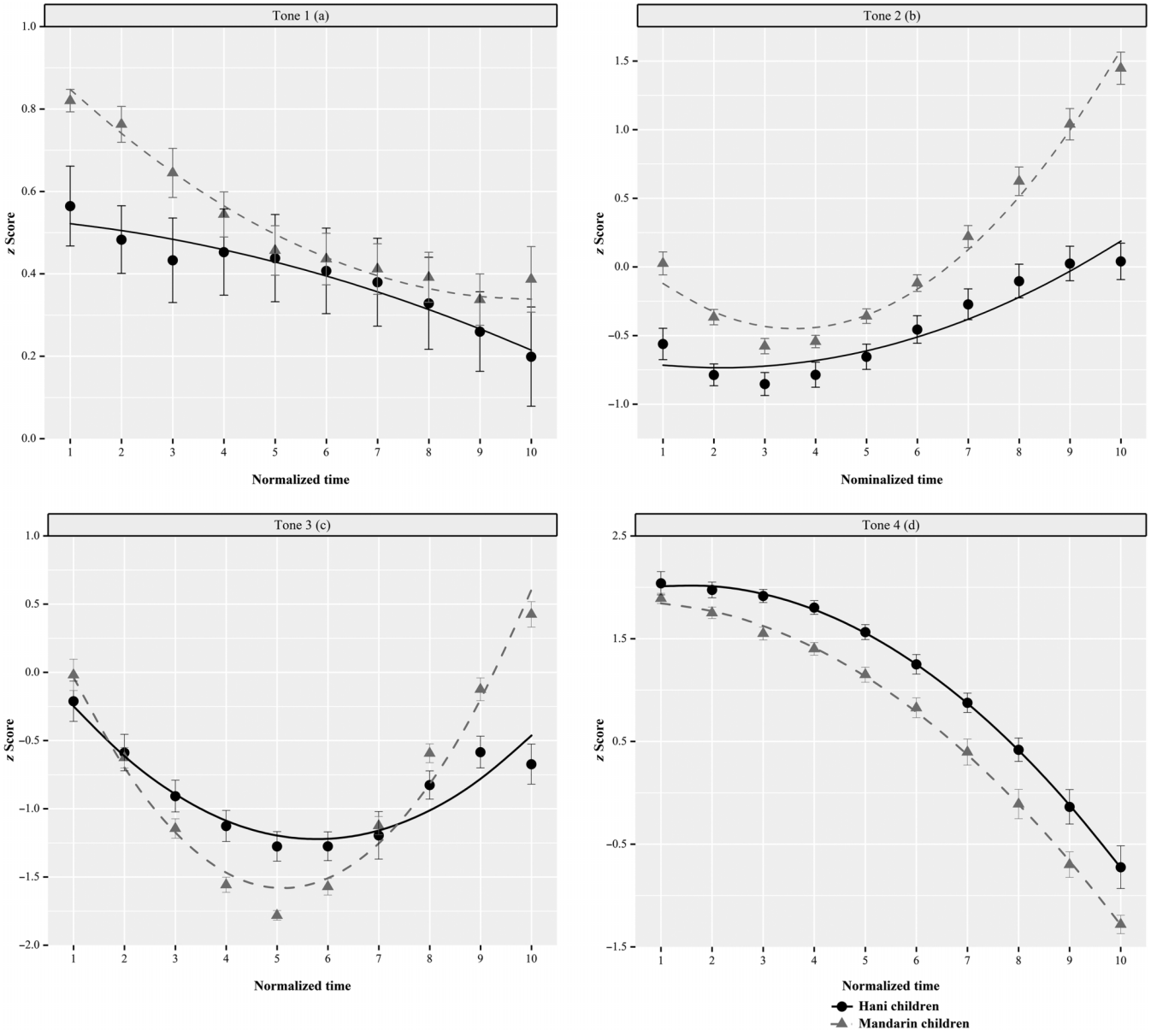


Figure 3b displays the pitch contours for T2 produced by both Hani–Mandarin bilingual children and Mandarin monolingual children. The analysis of the growth curve showed that the quadratic term, $\chi^2(1) = 31.715$, $p < .001$, and the slope, $\chi^2(1) = 5.149$, $p = .023$, were significant, but not the intercept, $\chi^2(1) = 1.631$, $p > .001$. This indicates that the pitch contours of the tones but not the pitch height produced by the Hani–Mandarin bilingual children differed significantly from those produced by the native Mandarin children.

Figure 3c illustrates the pitch contours for Mandarin T3 produced by Hani–Mandarin bilingual and Mandarin monolingual children. Both contours follow a similar trajectory from the onset to the midpoint. However, Mandarin monolingual children produce a steeper rising pitch from the midpoint to the offset than Hani–Mandarin bilingual children. Specifically, the Mandarin contour in the later portion of the syllable shows a steeper rising slope. In contrast, the Hani–Mandarin bilingual children's contours for T3 show a less steep pattern than their

Mandarin counterparts. The growth curve analysis revealed significant differences in the slope of the quadratic terms, $\chi^2(1) = 23.820$, $p < .001$. However, as indicated by the intercept, $\chi^2(1) = 2.867$, $p = .09$, the pitch height values were not significantly different from those of the Mandarin monolingual children.

Figure 3d displays the pitch contours of T4 produced by Hani–Mandarin bilingual children and Mandarin monolingual children. Both contours follow a similar trajectory from onset to offset. The analysis of the growth curve showed a significant difference in the pitch height, $\chi^2(1) = 9.537$, $p = .002$, but not in the quadratic, $\chi^2(1) = 0.895$, $p > .001$, and linear terms, $\chi^2(1) = 1.947$, $p > .001$, of the slope. This indicates that the pitch height of this tone, but not the *F*0 shape produced by the two groups, differed significantly.

Discussion

This study investigated the production of Mandarin tones by native Mandarin-speaking children and explored the challenges faced by young Hani-speaking children in acquiring these tones. Key challenges for the Hani children included differences between their L1 and Mandarin (L2) in tone categories, pitch height, pitch contour, as well as a bidirectional confusion between specific Mandarin tones. Overall, the Mandarin tones produced by Hani–Mandarin bilingual children showed notable deviations from those of native Mandarin children, differing in pitch contour, pitch height, or both. To our knowledge, this is the first study to analyze Mandarin tone production in native Hani learners, and the findings are discussed in detail below.

The Rising Tone Is More Difficult Than the Level and Falling Tones

The first research question examines which Mandarin tones are more challenging for Hani–Mandarin bilingual children. Our findings support the hypothesis that T2 and T3 pose greater difficulties than T1 and T4 for these children. Like their Mandarin monolingual peers, Hani–Mandarin bilingual children find T1 and T4 easier to produce. Despite minor acoustic differences in T1 and T4 between the two groups, these tones are generally perceived as accurate. In contrast, T2 and T3 not only differ acoustically but also exhibit higher error rates, particularly T3.

Consistent with our results, other studies have shown that bilingual children tend to acquire T1 and T4 more readily, while T2 and T3 present ongoing challenges (Fabra et al., 2021; Lin et al., 2025; Meckelborg et al., 2024). This pattern has been observed in children from both nontonal language backgrounds (L1 Spanish–L2 Mandarin, Fabra et al., 2021; L1 English–L2 Mandarin, Lin et al., 2025) and tonal backgrounds (L1 Cantonese–L2 Mandarin, Meckelborg et al., 2024). These findings suggest that while L2 learners may produce Mandarin tones in ways influenced by their L1, different L2 learner groups may also exhibit similar tonal errors (Winke, 2007). Why, then, are T1 and T4 more accessible to bilingual children? According to Fabra et al. (2021), tones with higher static pitch targets are articulatorily simpler, making them easier to produce for children whose native language does not rely on tones. In Mandarin, however, tones differ in both pitch height and pitch contour, which adds to their complexity.

Even among Mandarin monolinguals, T2 and T3 are the most distinct from adult forms, likely due to the intricate motor control required to produce these tones (Wong, 2012). Not all bilingual children struggle equally with these tones, however. For instance, Huang (2016) found that Min-Mandarin bilinguals can accurately produce Mandarin T1, T3, and T4, though they struggle with T2. Additionally, Fuzhou-Mandarin bilingual children exhibit high accuracy and similarity to native Mandarin speakers in tone production (You et al., 2024). Several factors may explain why Min-Mandarin and Fuzhou-Mandarin bilinguals produce Mandarin tones with such accuracy. The first factor contributing to successful Mandarin tone acquisition is the positive transfer from L1 to L2. In Min-Mandarin bilingual children, Mandarin tones are systematically assimilated into their native lexical tone categories. Specifically, Mandarin T1, T2, and T4 are mapped onto Min T1, T3, and T2, respectively. This systematic tone mapping facilitates Min-Mandarin bilingual children's accuracy in learning Mandarin tones. The second factor is the quantity of language exposure. Min-Mandarin bilingual children not only use Mandarin in school but also maintain continuous exposure to Mandarin in public domains. Among Fuzhou-Mandarin bilinguals, Mandarin has consistently been the dominant language in daily life, with its dominance increasing with age. In this group, Mandarin is used in 60% of home interactions, while the Fuzhou dialect accounts for only 40%. The combined effects of L1 positive transfer and substantial Mandarin exposure accelerate Mandarin tone acquisition in both bilingual groups. Notably, both Min-Mandarin and Fuzhou-Mandarin bilingual children are classified as simultaneous bilinguals, having been exposed to Mandarin from birth. This early exposure likely plays a critical role in their successful acquisition of Mandarin tones (Lin et al., 2025).

Furthermore, prior knowledge of a tonal language, despite differences between the Hani and Mandarin tonal systems, appears to aid Hani–Mandarin bilingual children

in producing Mandarin tones with relative accuracy. As noted in the introduction, Hani's high-level (T1) and low-falling tones (T3) align with Mandarin's high-level (T1) and high-falling tones (T4), facilitating a tonal mapping that aids acquisition. Our study confirms that positive transfer occurs when bilingual children apply phonetic patterns from one language to another. Conversely, negative transfer arises when features from one language interfere with the other (Fabiano-Smith & Goldstein, 2010). This highlights that negative transfer is especially common in early bilingual development as children adopt phonological traits from their dominant language (MacWhinney, 2005; Mayr et al., 2014).

Pitch Height and Pitch Contour Are Both Challenges

The second research question focused on pitch height and pitch contour. Our hypothesis that only pitch contour would present a challenge for Hani–Mandarin bilinguals was partially supported. Findings from this study indicate that both pitch height and pitch contour in Mandarin tones posed challenges for Hani–Mandarin bilingual children. Notably, these children did not achieve mastery in the pitch heights of T1 and T4, contrasting with results from prior research in which Spanish–Mandarin bilingual children succeeded in mastering these pitch heights but struggled with pitch contour (Fabra et al., 2021). This raises the following question: Why do Spanish–Mandarin bilingual children excel in mastering the pitch heights of T1 and T4, while Hani–Mandarin bilingual children face difficulties? We propose that these differences may relate to whether the bilingual child's native language is tonal. Research shows that nontonal language learners tend to focus on average pitch and onset/offset pitch height rather than pitch contour, while tonal language learners attend to both pitch height and pitch contour (So & Best, 2010; Wayland & Guion, 2004). Because Hani is a tonal language, it appears to exert a specific transfer effect in Hani–Mandarin bilinguals' acquisition of Mandarin. Although Hani contains level and falling pitch contours in its tonal system, its falling tones do not reach the higher pitch onsets required for Mandarin T4, which likely contributes to Hani–Mandarin bilingual children's difficulty in matching the pitch height of Mandarin speakers.

Furthermore, although Hani shares high-level tones with Mandarin, Hani–Mandarin bilinguals' production of Mandarin T1 still showed slight variations in pitch height compared to native Mandarin children. According to the SLM-r, bilingual learners may never fully match native speakers in their target L2 because the phonetic elements of their L1 and L2 systems interact, and their phonetic input in the L2 differs from that of native speakers (Flege & Bohn, 2021). Additionally, as outlined in the introduction, bilingual children typically operate within two language systems, which can lead to phonological patterns distinct from those of monolingual children (Lleó & Kehoe, 2002).

Moreover, studies on both Mandarin-speaking children and adults have shown that even when tones are judged as correct by adult evaluators in perceptual assessments, children's acoustic realizations still do not fully reach adultlike levels (Wong, 2012). Therefore, as Ladefoged and Disner (2012, p. 14) noted in their analysis of Mandarin tones:

> We cannot say that a particular tone has certain frequency values. ... The absolute values are never important from a linguistic point of view. The changes in meaning are conveyed by the relative pitch changes. The pitch range used is a property of the speaker, not the language.

Our growth curve analyses further showed that Hani–Mandarin bilinguals did not master the pitch contours of Mandarin T2 and T3, in line with prior findings (Fabra et al., 2021; Meckelborg et al., 2024). Notably, Meckelborg et al. (2024) observed that while Cantonese–Mandarin bilinguals could produce T3 correctly, their T3 production lacked the refinement of native Mandarin speakers. Meanwhile, Spanish–Mandarin bilinguals failed to produce any native-like Mandarin pitch contours (Fabra et al., 2021). Specifically, Hani–Mandarin bilinguals' rising portion in T2 was insufficient, and the dipping point of T3 was unclear (see Figure 2).

These results may be attributed to both native language transfer effects and the role of laryngeal muscle activity related to *F*0. Studies consistently show that L1 impacts both production and perception in L2 phonology (Bergmann et al., 2016; Dmitrieva et al., 2010; Flege, 2003; Kong & Kang, 2023; Lang & Davidson, 2019; MacKay et al., 2001; Trofimovich et al., 2007). Notably, the tonal system of Hani does not include tones with a rising contour. In other words, Hani–Mandarin bilingual children's L1 lacks tones that combine both falling and rising contours within a single tone, thereby precluding positive transfer for T2 and T3.

From a physiological standpoint, Hallé (1994) found that the cricothyroid muscle is crucial in raising *F*0, while the sternohyoid muscle governs *F*0 decline. The cricothyroid muscle primarily influences T2 and T3, tones that involve complex rises and falls in pitch. Fabra et al. (2021) suggested that Mandarin T3 is the most complex tone due to these intricate muscle movements, followed by T2. Coordinated motor actions needed for both falling

and rising pitch may present a greater challenge than tones requiring a single movement (Wong, 2012). Consistent with these findings, children struggle more with T3 regardless of language background (Meckelborg et al., 2024). Studies on Mandarin-speaking children's acquisition of tones also show that T2 and T3 are harder to master than T1 and T4 (Li & Thompson, 1977; Wong, 2012; Wong et al., 2005). Mandarin-speaking children require the most time to acquire T3 (Wong, 2012, 2013), as its defining low pitch is generally more difficult to articulate (Xu, 1997). Tones with more complex *F*0 contours are particularly challenging for children to master (Wong & Strange, 2017), making it unsurprising that Hani–Mandarin bilingual children fail to produce the corresponding pitch contours.

The Bidirectional Confusion Between T2 and T3

Our findings indicate that Hani–Mandarin bilingual children sometimes produce T3 as T2 or T2 as T3, aligning with previous research documenting bidirectional confusion between T2 and T3 (Hao, 2012; Lin et al., 2025; Meckelborg et al., 2024). Growth curve analysis revealed no significant difference in pitch height between T2 and T3 in these bilingual children, suggesting they may not differentiate these tones based on pitch height alone but rather rely on pitch contour. Given the shared characteristics of T2 and T3—both involving falling and rising contours with similar onset pitch heights—such confusion is unsurprising. Previous studies suggest that this confusion may stem from general phonetic similarity (Meckelborg et al., 2024), the intrinsic difficulty of the T2–T3 distinction (Hao, 2012), or the simplification and complication of pitch targets (Lin et al., 2025), rather than the influence of the native language. Indeed, research indicates that both Mandarin-speaking children and adults frequently confuse T2 and T3 in perception and production due to their similar rising contours (Wong, 2012; Wong et al., 2005).

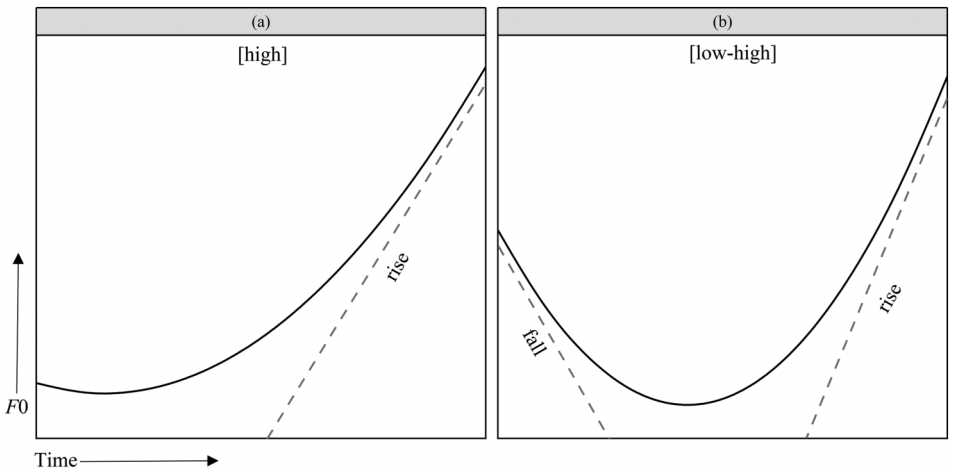
Limited early exposure to Mandarin may reduce Hani–Mandarin bilingual children's sensitivity to the specific pitch contours of T2 and T3, leading them to develop phonetic representations distinct from those of monolingual Mandarin speakers (Kuhl et al., 2008). Furthermore, the SLM-r posits that tone substitution from the L1 is common among early L2 learners (Flege & Bohn, 2021). This suggests that even as these bilingual children progress in their Mandarin acquisition (e.g., by sixth grade), their tonal development may still bear markers of early L2 acquisition.

Empirical studies also reveal that even at advanced stages of L2 learning, bilinguals often produce sound categories that, while acoustically distinct from L1, do not fully match monolingual L2 articulations. This occurs because sound categories from both languages continue to influence one another (Bergmann et al., 2016; Dmitrieva et al., 2010; Flege, 2003; Guion, 2003; Lang & Davidson, 2019; MacKay et al., 2001). Therefore, the overlap between Hani and Mandarin tonal categories is likely a persistent feature in the Mandarin acquisition process of Hani–Mandarin bilingual children.

Limitations and Future Directions

Our study has certain limitations that future research could address. Mandarin tones differ between monosyllabic and disyllabic words, with some tones undergoing changes

Figure 4. A schematic illustration of the hypothetical surface *F*0 realization (solid line) and pitch target (dotted lines): (a) the hypothetical surface *F*0 (solid curved line) and a single static target (dotted line); (b) the hypothetical surface *F*0 (solid curved line) and a single dynamic target (dotted line). *F*0 = fundamental frequency.



in disyllabic contexts. For instance, the phonetic realization of Mandarin T3 is highly variable (Chang & Yao, 2016; Hallé et al., 2004). In specific contexts, T3 may be realized as a low-level rising tone or even as T2 when directly followed by another T3, often taking on a creaky quality (Chang & Yao, 2016; Lin, 1985). This variability can make accurate T3 production and perception challenging, increasing the potential for confusion with T2. Future studies could explore how bilingual and monolingual children process these tones in disyllabic words, providing insights into potential differences in their ability to coordinate tone production. Additionally, research shows that L2 Mandarin learners generally perform well in producing isolated lexical tones (Hao, 2012; Wang et al., 2003). Future studies could further investigate the accuracy of Mandarin tone production among L2 learners in both monosyllabic and disyllabic contexts. This approach would allow for a more precise and comprehensive analysis of L2 tone acquisition.

This study also has practical and pedagogical implications. In Mandarin instruction, teachers often overlook students' linguistic backgrounds, resulting in teaching methods that may not be suitable for all learners. Mandarin tones are fundamental to early language acquisition, characterized by pitch height and pitch contour. Our findings highlight that the rising pitch contour of Mandarin tones presents a particular challenge for Hani-speaking children. Therefore, special attention should be given to pitch contour when teaching Mandarin to Hani-speaking children. Xu and Wang (2001, p. 321) proposed that Mandarin Chinese tones consist of two components: the surface *F*0 contour and the underlying pitch target. According to their framework, the *F*0 contour itself is not a linguistic unit, but rather the surface realization of the tonal category as a functional linguistic unit. Figure 4 provides a schematic representation of the relationship between surface *F*0 realization and pitch targets. Xu and Wang identified two fundamental types of pitch targets: static and dynamic. A static pitch target consists of a single pitch target, such as Mandarin T2, which has an underlying (high) pitch target. In contrast, a dynamic pitch target consists of two pitch targets. For instance, Mandarin T3 in isolation follows a dynamic pattern, composed of either a (low) target followed by a (mid) target or a (low) target followed by a (high) target. The difficulty experienced by Hani–Mandarin bilingual children in producing T2 and T3 pitch contours likely stems from a lack of awareness of their underlying pitch targets. Therefore, an instructional approach based on pitch targets could be beneficial. Specifically, for T2, pedagogical interventions could incorporate a rising gesture that aligns with its high pitch target. Similarly, for T3, instructional strategies could employ a falling-then-rising gesture, corresponding to its low-high pitch target sequence. These insights can help Mandarin teachers better consider the linguistic backgrounds of bilingual children and adopt targeted strategies when teaching Mandarin to children from diverse language backgrounds.

Conclusions

The present study examined Hani–Mandarin bilingual children's production of Mandarin tones. In Mandarin tones production, we found that the degree of difficulty in production varied across Mandarin tones: Hani–Mandarin bilingual children could master Mandarin T1 and T4 pitch contours but not pitch heights; they could master T2 and T3 pitch heights but not pitch contours. Our observations indicate that tone experience in L1 not only facilitates or hinders tone acquisition in an L2—highlighting both the advantages and disadvantages of native tone patterns for nonnative lexical tone production—but also that physiological factors, such as the activation patterns of laryngeal muscles, play a significant role in tone acquisition. We suggest further research on the development of tone production and perception in L1 and L2, as Mandarin tone learning is lengthy (Wong, 2013; Wong & Strange, 2017).

Data Availability Statement

The data sets used for this study are available from the corresponding author on reasonable request.

Acknowledgments

This study did not receive any funding. The authors would also like to thank the participating schools, principals, teachers, children, and parents for their support; Yufen Wu and Yuexuan Li for their assistance with data collection and transcription of parts of the recordings; and special thanks to Liming Zhao for his help in preparing for the research.

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Appendix A

Demographic Background Information

Table A1. The summary demographics of participants.

Demographic	Hani bilinguals (n = 30)			Native speakers (n = 20)		
	M	***SD***	**Range**	***M***	***SD***	**Range**
Age	11.00	0.78	10–12	10.9	0.87	10–12
Age of acquisition						
Chinese	3.73	0.98	2–6	—	—	—
Self-ratings Chinese	7.80	0.26	7–9	8.97	0.23	8–10
Understanding	7.50	1.11	6–9	9.1	0.73	8–10
Speaking	8.00	0.69	7–9	8.7	0.67	8–10
Hearing	7.90	0.66	7–9	9.1	0.87	8–10

Note. The maximum score was 10 for self-rated proficiency. Em dashes indicate "not applicable."

Appendix B

Materials

Table B1. The recording words used in this study.

Tone	1	2	3	4	5	6
Tone 1	八bā "eight"	低dī "low"	巴bā "wish"	搭dā "build"	逼bī "force"	都dū "capital"
Tone 2	答dá "answer"	读dú "read"	敌dí "enemy"	鼻bí "nose"	拔bá "pull"	达dá "reach"
Tone 3	把bǎ "grip"	笔bǐ "pen"	打dǎ "hit"	比bǐ "compare"	底dǐ "bottom"	堵dǔ "block"
Tone 4	爸bà "dad"	布bù "fabric"	大dà "big"	弟dì "brother"	肚dù "belly"	地dì "groud"

Appendix C

Confusion Matrix Data

Table C1. Confusion matrices of Hani–Mandarin children' production of citation tones.

Target tone	Transcribed production					
	Tone 1	**Tone 2**	**Tone 3**	**Tone 4**	**Semi-Tone 3**	**Uncategorizable**
Tone 1	—	0 0.0%	0 0.0%	2 25%	0 0.0%	**6 75%**
Tone 2	0 0.0%	—	**14 77.8%**	0 0.0%	4 22.2%	0 0.0%
Tone 3	0 0.0%	**28 37.8%**	—	0 0.0%	**46 62.2%**	0 0.0%
Tone 4	0 0.0%	0 0.0%	0 0.0%	—	0 0.0%	**2 100%**

Note. The cell was bolded if the percentage of this pattern exceeded 30% among all patterns for this target.

Appendix D (p. 1 of 3)

Statistical Model Results

Table D1. Summary of the final statistical model with the Mandarin children group as the reference level in Tone 1.

Parameter	**Estimate**	***SE***	***t***	***p***
Fixed effects				
Intercept	0.394	0.074	5.308	.000
Linear	–0.310	0.061	–5.057	.000
Quadratic	–0.050	0.068	–0.737	.461
Hani children: intercept	0.124	0.115	1.075	.282
Hani children: linear	–0.204	0.095	–2.146	.032
Hani children: quadratic	0.194	0.105	1.849	.065
Random effects (intercept)		**Random slopes**	**Variance**	***SD***
Participants			0.077	0.277
		Linear	0.048	0.219
		Quadratic	0.060	0.244
Information criteria		**Estimate**		
Log-likelihood		191.10		
Deviance information criterion		–382.20		
Akaike information criterion		–356.20		
Bayesian information criterion		–311.00		
R^2		**Estimate**		
Marginal[a]		.18		
Conditional[b]		.96		

[a]Marginal R^2 describes the proportion of variance explained by the fixed factors alone. [b]Conditional R^2 describes the proportion of variance explained by both the fixed and random factors.

Table D2. Summary of the final statistical model with the Mandarin children group as the reference level in Tone 2.

Parameter	**Estimate**	***SE***	***t***	***p***
Fixed effects				
Intercept	–0.442	0.076	–5.854	.000
Linear	0.913	0.120	7.608	.000
Quadratic	0.341	0.063	5.430	.000
Hani children: intercept	0.580	0.117	4.960	.000
Hani children: linear	0.804	0.186	4.325	.000
Hani children: quadratic	0.791	0.097	8.123	.000
Random effects (intercept)		**Random slopes**	**Variance**	***SD***
Participants			0.078	0.279
		Linear	0.178	0.422
		Quadratic	0.032	0.179
Information criteria		**Estimate**		
Log-likelihood		34.40		
Deviance information criterion		–68.80		
Akaike information criterion		–42.80		
Bayesian information criterion		2.50		
R^2		**Estimate**		
Marginal[a]		.72		
Conditional[b]		.94		

[a]Marginal R^2 describes the proportion of variance explained by the fixed factors alone. [b]Conditional R^2 describes the proportion of variance explained by both the fixed and random factors.

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Statistical Model Results

Table D3. Summary of the final statistical model with the Mandarin children group as the reference level in Tone 3.

Parameter	**Estimate**	***SE***	***t***	***p***
Fixed effects				
Intercept	–0.867	0.056	–15.440	.000
Linear	–0.216	0.192	–1.127	.260
Quadratic	0.979	0.113	8.628	.000
Hani children: intercept	0.055	0.087	0.628	.530
Hani children: linear	0.864	0.297	2.908	.004
Hani children: quadratic	1.122	0.176	6.384	.000
Random effects (intercept)		**Random slopes**	**Variance**	***SD***
Participants			0.037	0.193
		Linear	0.447	0.668
		Quadratic	0.112	0.335
Information criteria		**Estimate**		
Log-likelihood		–74.30		
Deviance information criterion		148.50		
Akaike information criterion		174.50		
Bayesian information criterion		219.80		
R^2		**Estimate**		
Marginal[a]		.62		
Conditional[b]		.84		

[a]Marginal R^2 describes the proportion of variance explained by the fixed factors alone. [b]Conditional R^2 describes the proportion of variance explained by both the fixed and random factors.

Table D4. Summary of the final statistical model with the Mandarin children group as the reference level in Tone 4.

Parameter	**Estimate**	***SE***	***t***	***p***
Fixed effects				
Intercept	1.098	0.072	15.285	.000
Linear	–2.765	0.192	–14.434	.000
Quadratic	–0.876	0.064	–13.696	.000
Hani children: intercept	–0.411	0.111	–3.691	.000
Hani children: linear	–0.392	0.297	–1.320	.187
Hani children: quadratic	0.095	0.099	0.955	.340
Random effects (intercept)		**Random slopes**	**Variance**	***SD***
Participants			0.0713	0.267
		Linear	0.505	0.710
		Quadratic	0.048	0.220
Information criteria		**Estimate**		
Log-likelihood		106.20		
Deviance information criterion		–212.30		
Akaike information criterion		–186.3		
Bayesian information criterion		–141.10		
R^2		**Estimate**		
Marginal[a]		.88		
Conditional[b]		.99		

[a]Marginal R^2 describes the proportion of variance explained by the fixed factors alone. [b]Conditional R^2 describes the proportion of variance explained by both the fixed and random factors.

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